

Program 7-Deploy a Web Application to Kubernetes using Minikube

Directory structure:

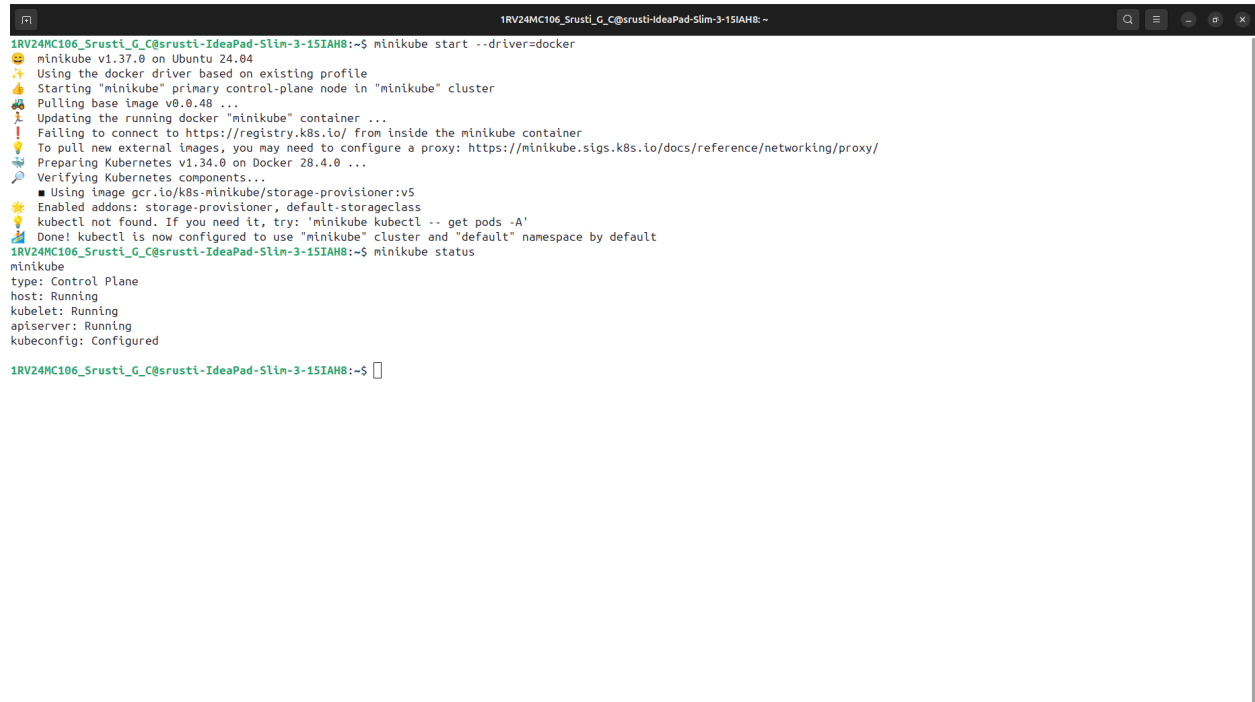
— Program7

— nginx-deployment.yaml

Step 1: Start Minikube

minikube start

minikube status

A terminal window titled '1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slim-3-15IAH8: ~' showing the execution of 'minikube start --driver=docker' and 'minikube status'. The start command output includes: 'minikube v1.37.0 on Ubuntu 24.04', 'Using the docker driver based on existing profile', 'Starting "minikube" primary control-plane node in "minikube" cluster', 'Pulling base image v0.0.48 ...', 'Updating the running docker "minikube" container ...', 'Failing to connect to https://registry.k8s.io/ from inside the minikube container', 'To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/', 'Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...', 'Verifying Kubernetes components...', 'Using image gcr.io/k8s-minikube/storage-provisioner:v5', 'Enabled addons: storage-provisioner, default-storageclass', 'kubectl not found. If you need it, try: "minikube kubectl -- get pods -A"', and 'Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default'. The status command output shows: 'minikube', 'type: Control Plane', 'host: Running', 'kubernetes: Running', 'apiserver: Running', and 'kubeconfig: Configured'.

```
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slim-3-15IAH8:~$ minikube start --driver=docker
minikube v1.37.0 on Ubuntu 24.04
Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.48 ...
Updating the running docker "minikube" container ...
Failing to connect to https://registry.k8s.io/ from inside the minikube container
To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Verifying Kubernetes components...
  ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
  ■ Enabled addons: storage-provisioner, default-storageclass
  ■ kubectl not found. If you need it, try: "minikube kubectl -- get pods -A"
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slim-3-15IAH8:~$ minikube status
minikube
type: Control Plane
host: Running
kubernetes: Running
apiserver: Running
kubeconfig: Configured
```

Step 2: Create deployment file

nano nginx-deployment.yaml

Step 3: Apply deployment

kubectl apply -f nginx-deployment.yaml

kubectl get deployments

kubectl get pods

```
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~$ kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
kubia                0/1     ContainerCreating   0           18s
nginx-deployment-6f9664446b-48ffdf  1/1     Running        1 (11d ago)  11d
nginx-deployment-6f9664446b-nfcxw  1/1     Running        1 (11d ago)  11d
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~$ kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
kubia                0/1     ContainerCreating   0           26s
nginx-deployment-6f9664446b-48ffdf  1/1     Running        1 (11d ago)  11d
nginx-deployment-6f9664446b-nfcxw  1/1     Running        1 (11d ago)  11d
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~$ kubectl expose pod kubia --type=NodePort --port=8080
service/kubia exposed
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~$ kubectl get svc
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1     <none>         443/TCP           12d
kubia                NodePort    10.107.157.15 <none>         8080:32430/TCP   12s
nginx-deployment    NodePort    10.97.222.48 <none>         80:31057/TCP     11d
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~$ minikube ip
192.168.49.2
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~$ http://192.168.49.2:32430
bash: http://192.168.49.2:32430: No such file or directory
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~$ curl http://192.168.49.2:32430
curl: (7) Failed to connect to 192.168.49.2 port 32430 after 0 ms: Couldn't connect to server
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~$ curl http://192.168.49.2:32430
curl: (7) Failed to connect to 192.168.49.2 port 32430 after 0 ms: Couldn't connect to server
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~$ minikube ip
192.168.49.2
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~$ kubectl get svc
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1     <none>         443/TCP           12d
kubia                NodePort    10.107.157.15 <none>         8080:32430/TCP   8m18s
nginx-deployment    NodePort    10.97.222.48 <none>         80:31057/TCP     11d
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~$ curl http://192.168.49.2:32430
curl: (7) Failed to connect to 192.168.49.2 port 32430 after 0 ms: Couldn't connect to server
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~$ curl http://192.168.49.2:32430
You've hit kubia
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-15IAH8:~$
```

Step 4:Expose deployment

kubectl expose deployment nginx-deployment --type=NodePort --port=80

Step 5:Access app

kubectl get services

minikube service nginx-deployment

  http://localhost:8080

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.